Lab 4 Arduino Sensor with WiFi Communication

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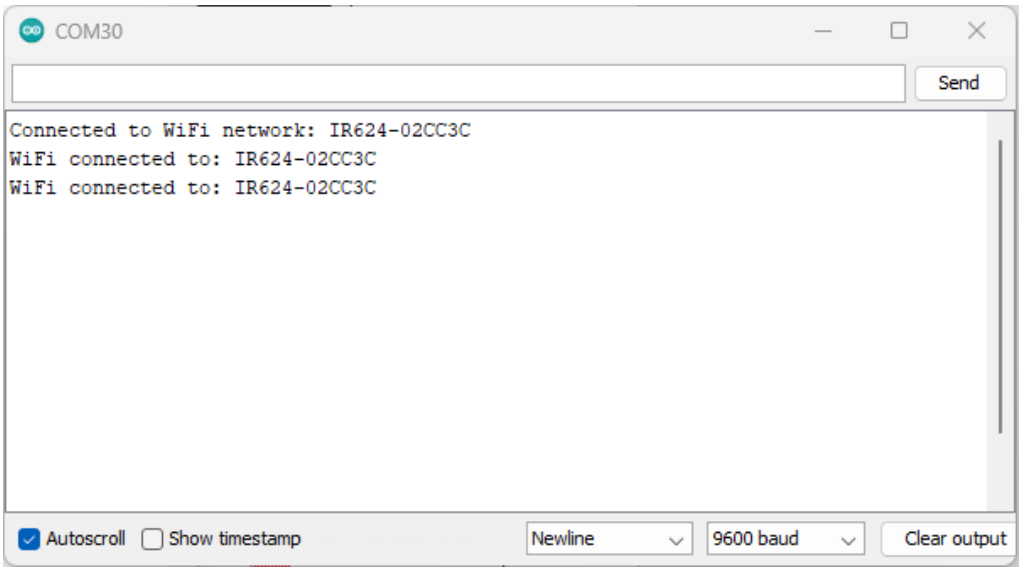
By default, the Arduino UNO R4 has an onboard WiFi communication module based on ESP32-S3.

1. Test the WiFi network connection

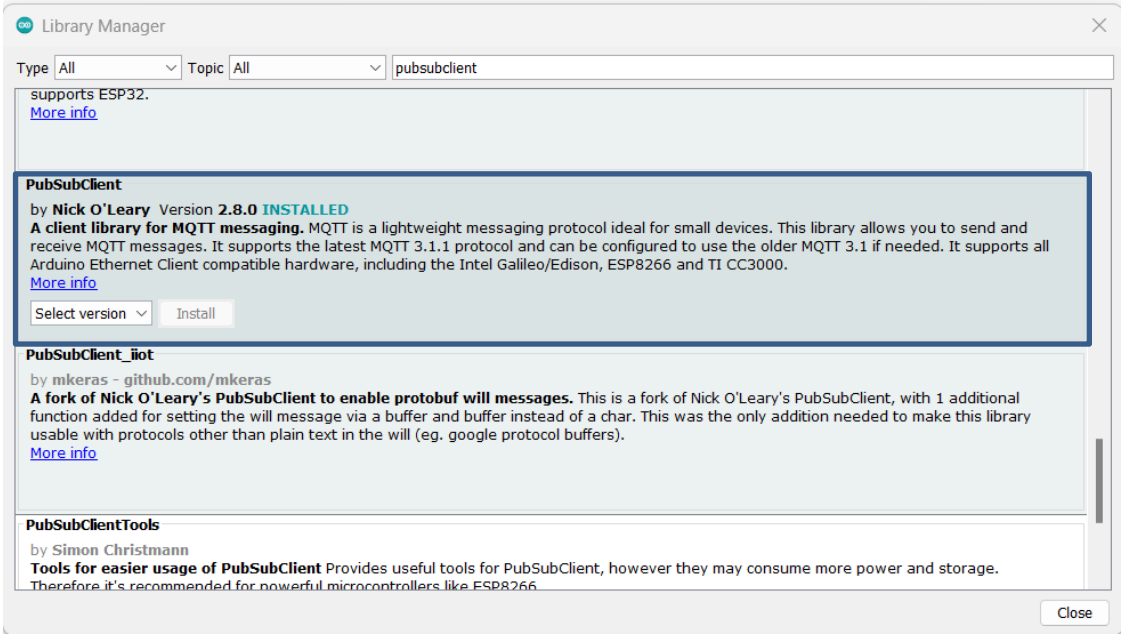
First, let us test if our Arduino UNO R4 WIFI can be connected to the WiFi Network. The following code can be used to test the WiFi connection. Compile the code, upload it to the Arduino board, and check the Wifi network connection status through the Serial Monitor.

```
#include <WiFiS3.h>
const char* ssid = "";  // SSID of the WiFi network
const char* password = "";  // WiFi password

void setup() {
  Serial.begin(9600);                        // Enable serial interface
  WiFi.begin(ssid, password);                // Connect to WiFi network
  while (WiFi.status() != WL_CONNECTED) {    // Check the status of
connection
    delay(500);
    Serial.println("Connecting to WiFi...");
  }
  Serial.print("Connected to WiFi network: ");
  Serial.println(ssid);
}

void loop() {
  //Check WiFi connection status
  if(WiFi.status() == WL_CONNECTED) {
    Serial.print("WiFi connected to: ");
    Serial.println(ssid);
  } else {
    Serial.println("WiFi disconnected");
  }
  delay(10000); // Wait for 10 seconds before checking again
}
```



2. Arduino to ThingSpeak with MQTT

In previous labs, we learned how to send data from Raspberry PI to ThingSpeak using the MQTT protocol. With Arduino, it is possible to send MQTT data. To do so, we will need an Arduino library called PubSubClient. Install it through the Arduino Library manager.



After the pubsubclient library has been installed, refer to the following code, which randomly generates two values and sends them to ThingSpeak with MQTT. Remember to set the WiFi network and the ThingSpeak channel parameters currently, as we did before. Moreover, as we are going to send two sensor values to ThingSpeak, we will need to enable two fields, namely “Field 1” & “Field 2” (see below). In this example, sensorValue1 is published to “Field 1” and sensorValue2 is published to “Field 2”.

Channel Settings

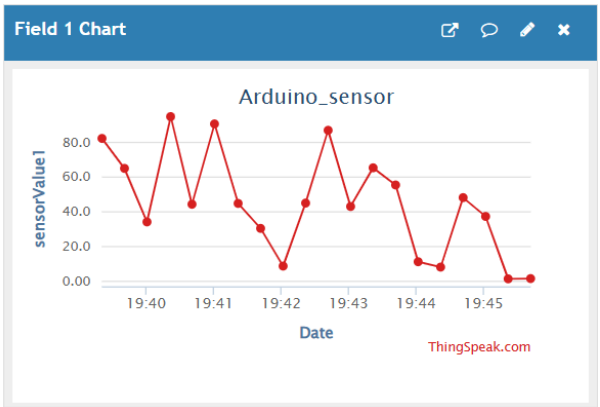
Percentage Complete	50%
Channel ID	1617746
Name	Arduino_sensor
Description	Using Arduino UNO WiFi to send temperature values to ThingSpeak platform
Field 1	Sensor1 ☑
Field 2	Sensor2 ☑
Field 3	☐
Field 4	☐

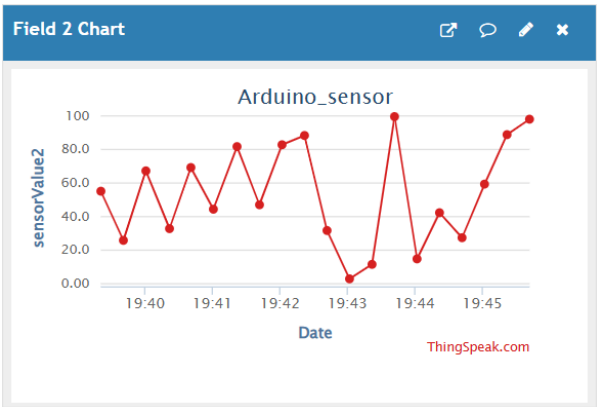
```
#include <WiFiS3.h>
#include <PubSubClient.h>

// Wi-Fi ssid and password
const char* ssid = "";
const char* password = "";

// MQTT broker
char *server = "mqtt3.thingspeak.com";
char *clientID = "";
char *topic = "";
char *userName = "";
char *mqttPassword = "";

// Declare WiFi and MQTT client
WiFiClient UNOR4Client;
PubSubClient client(server, 1883, NULL, UNOR4Client);

void startWifi() {
  Serial.print("\nUNOR4 connect with ");
  Serial.println(ssid);
  while (WiFi.status() != WL_CONNECTED)
  {
    Serial.println("Starting wifi connect...");
    WiFi.begin(ssid, password);
```

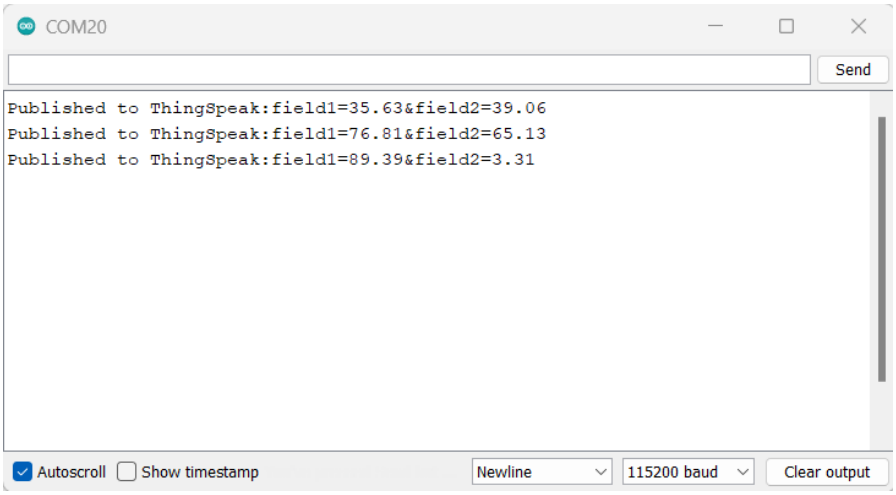
```
    delay(2000);
  }
  Serial.println("Wifi connected!");
  Serial.print("UNOR4 IP address: ");
  Serial.println(WiFi.localIP());
}

void connectBroker() {
  // User and password configured in the broker
  if (client.connect(clientID, userName, mqttPassword)) { //clientID,
username, password
    Serial.println("ThingSpeak connected!");
  }
}

void setup()
{
  Serial.begin(115200);
  startWifi();
  connectBroker();
}

unsigned long previousMillis = 0;
const long interval = 20000;  // 20 seconds
void loop()
{
  if (!client.connected()) {
    Serial.println("ThingSpeak disconnected, reconnecting ...");
    connectBroker();  // If not connected, attempt to reconnect
    delay(5000);
  }
  else{
    client.loop();
    unsigned long currentMillis = millis();
    // Run only if interval has passed
    if (currentMillis - previousMillis >= interval) {
      previousMillis = currentMillis;
      float sensorValue1 = random(0,10000)/100.0;
      float sensorValue2 = random(0,10000)/100.0;
      char msg[50];
      sprintf(msg, "field1=%.2f&field2=%.2f", sensorValue1,
sensorValue2);
      client.publish(topic, msg);
      Serial.print("Published to ThingSpeak:");
```

```
      Serial.println(msg);
    }
  }
}
```

If everything goes well, you will see the data published to ThingSpeak channel.



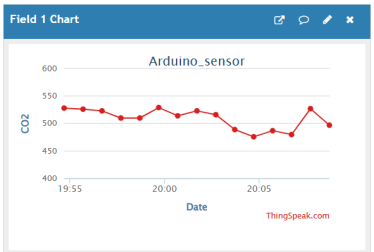


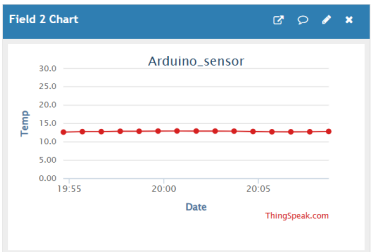


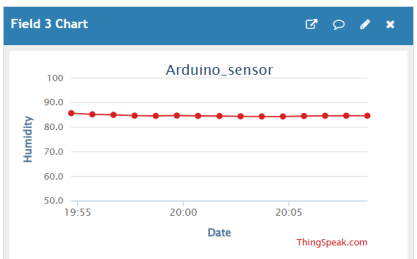
Practice task:

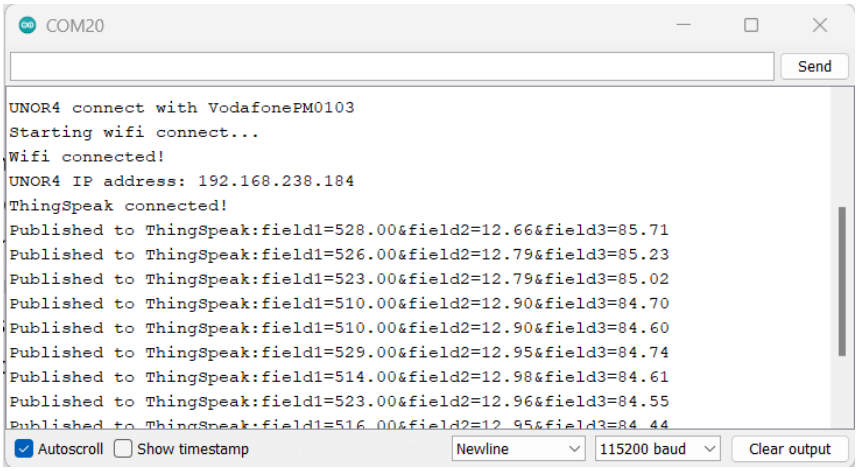
In previous labs, we have learned how to monitor environmental parameters with the SCD4x sensor, including CO2, temperature, and humidity. With MQTT communication, it is possible to send all these data to ThingSpeak. Design the Arduino code to do so. Here are the requirements:

- Measure the environmental parameters (CO2, temp, and humidity) with SCD4x every minute
- Publish the data to ThingSpeak with MQTT and visualize them in three figures
- Document the process and submit it together with the code








COM20
Send
UNOR4 connect with VodafonePM0103
Starting wifi connect...
Wifi connected!
UNOR4 IP address: 192.168.238.184
ThingSpeak connected!
Published to ThingSpeak:field1=528.00&field2=12.66&field3=85.71
Published to ThingSpeak:field1=526.00&field2=12.79&field3=85.23
Published to ThingSpeak:field1=523.00&field2=12.79&field3=85.02
Published to ThingSpeak:field1=510.00&field2=12.90&field3=84.70
Published to ThingSpeak:field1=510.00&field2=12.90&field3=84.60
Published to ThingSpeak:field1=529.00&field2=12.95&field3=84.74
Published to ThingSpeak:field1=514.00&field2=12.98&field3=84.61
Published to ThingSpeak:field1=523.00&field2=12.96&field3=84.55
Autoscroll
Show timestamp
Newline
115200 baud
Clear output